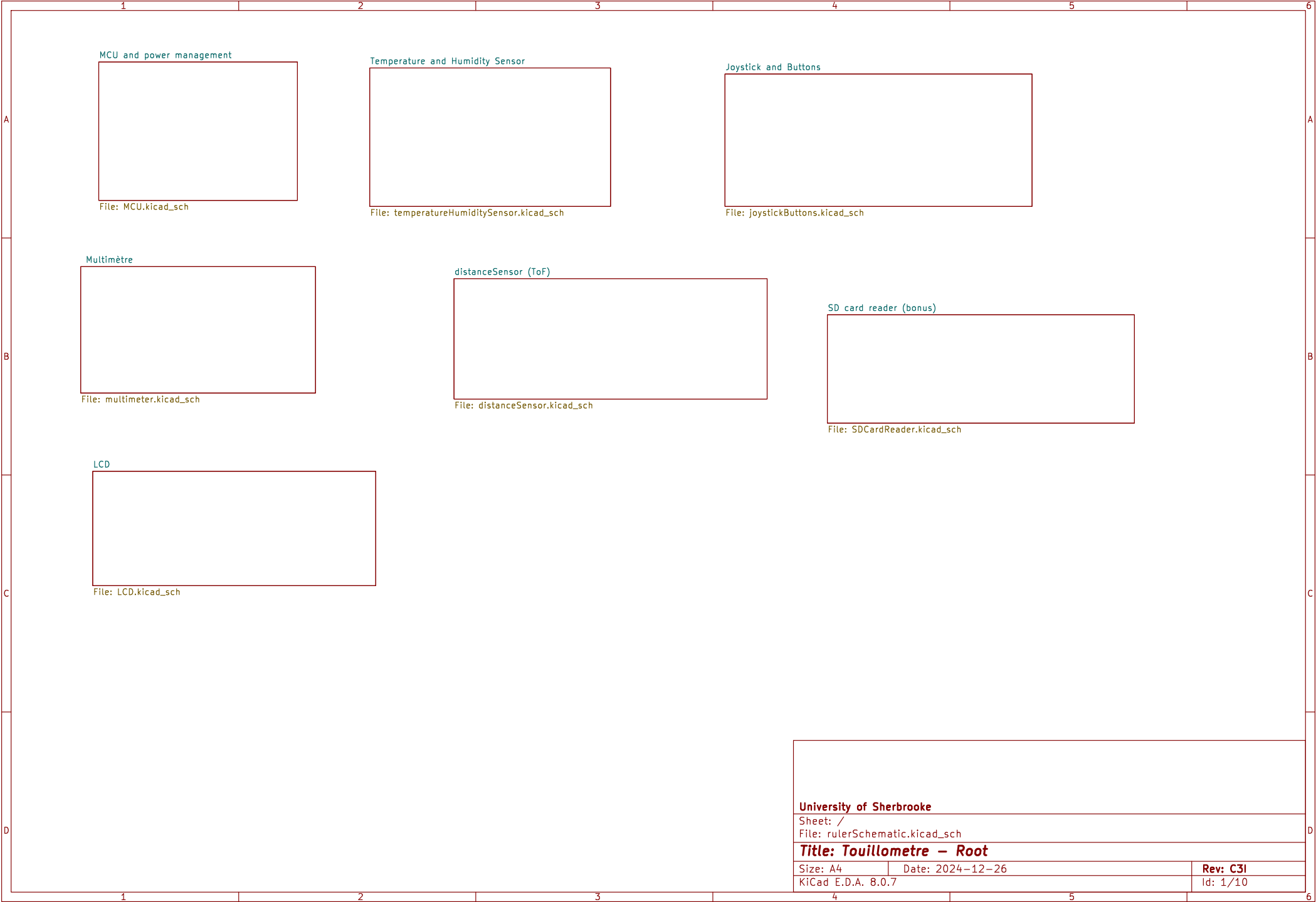
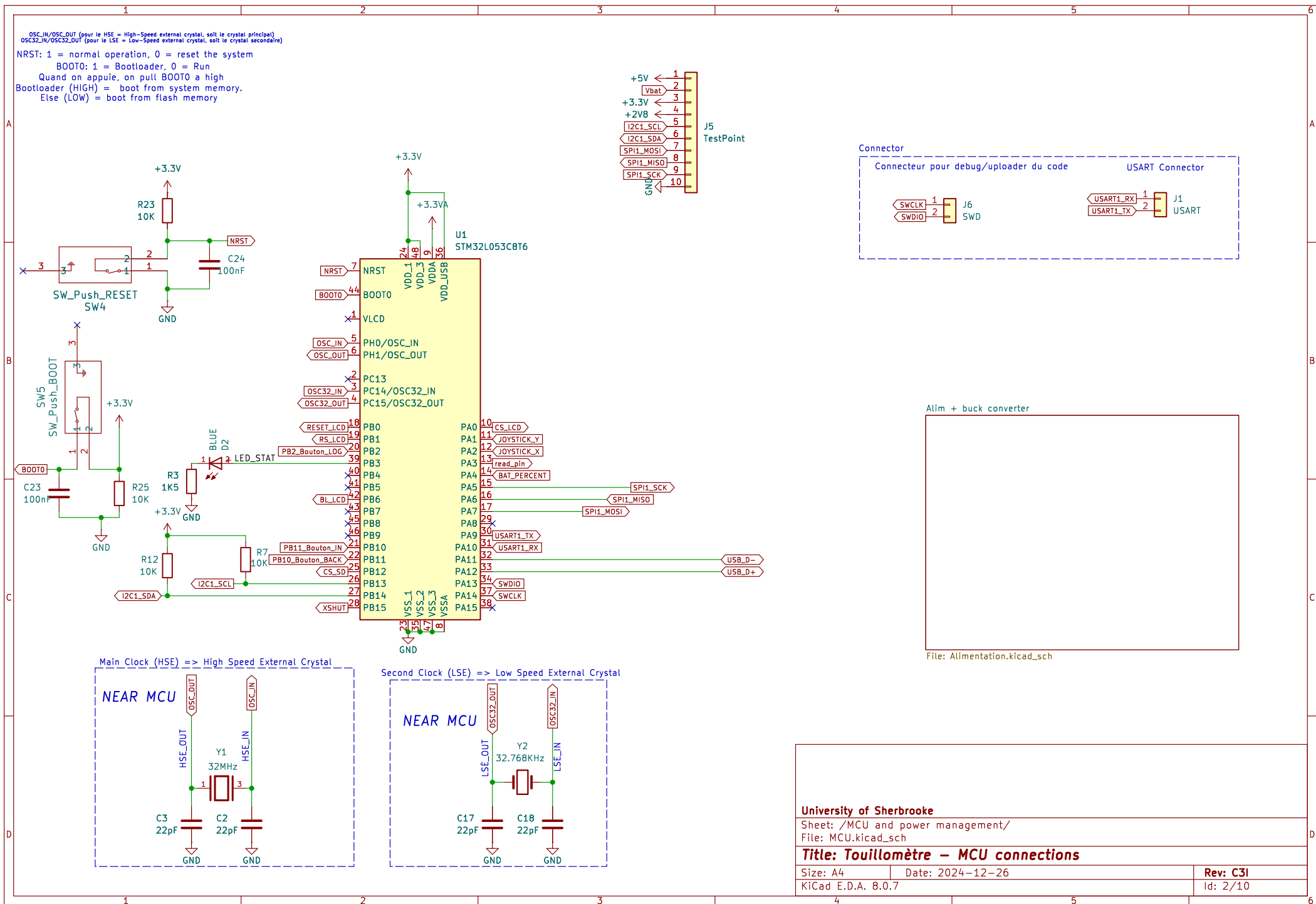
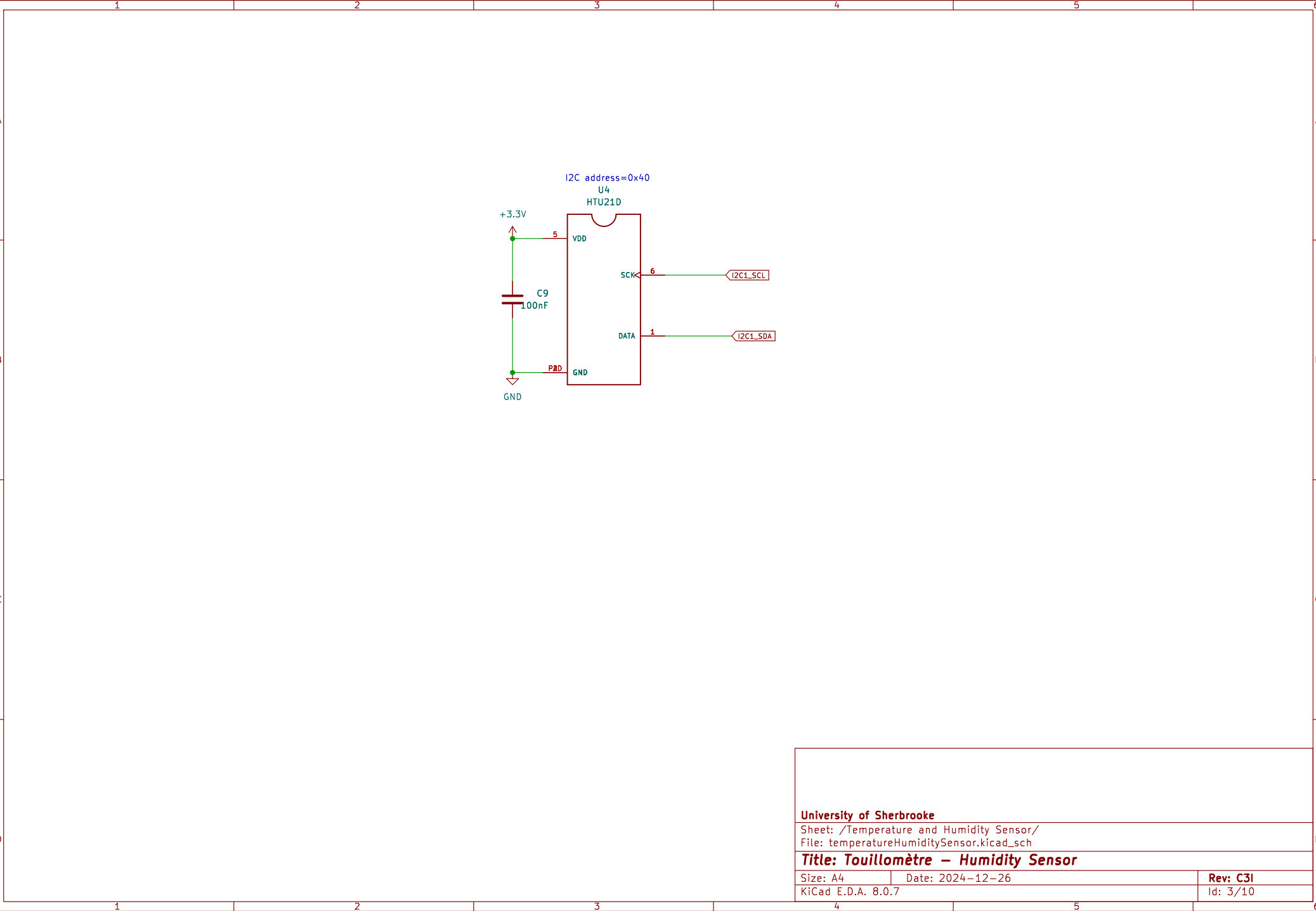


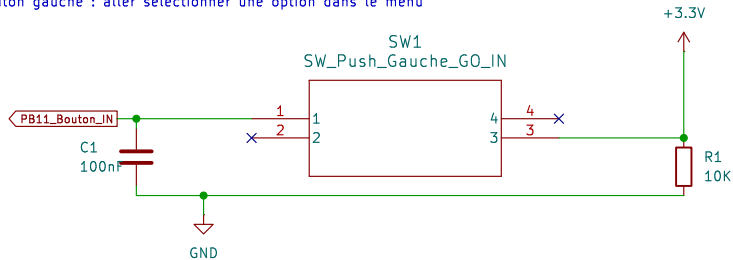




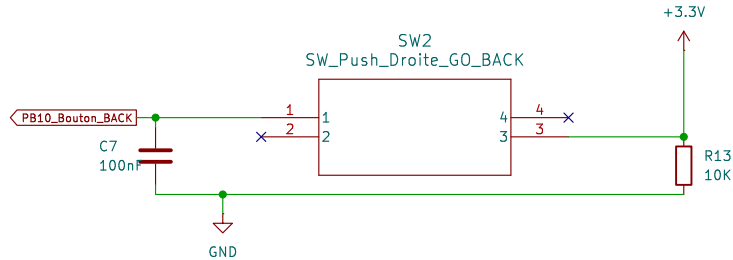


Exemple de pull-down qu'on devrait suivre (PULL-DOWN)

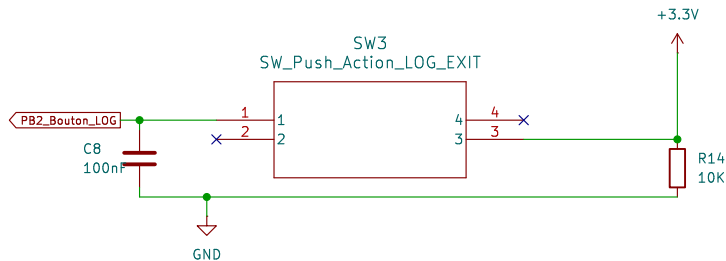
bouton gauche : aller sélectionner une option dans le menu



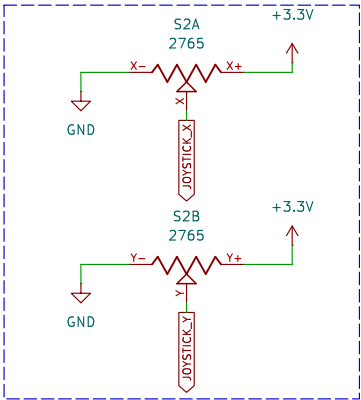
Bouton droit : aller revenir d'une option (retourner dans le menu)



click "log info" (ex: valeur de temp, distance...) / Exit magouille



Thumb joystick Adafruit (2765). Il n'y a pas de bouton



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Sheet: /Joystick and Buttons/

File: joystickButtons.kicad\_sch

Title: Touillomètre – Joystick and Buttons

Size: A4

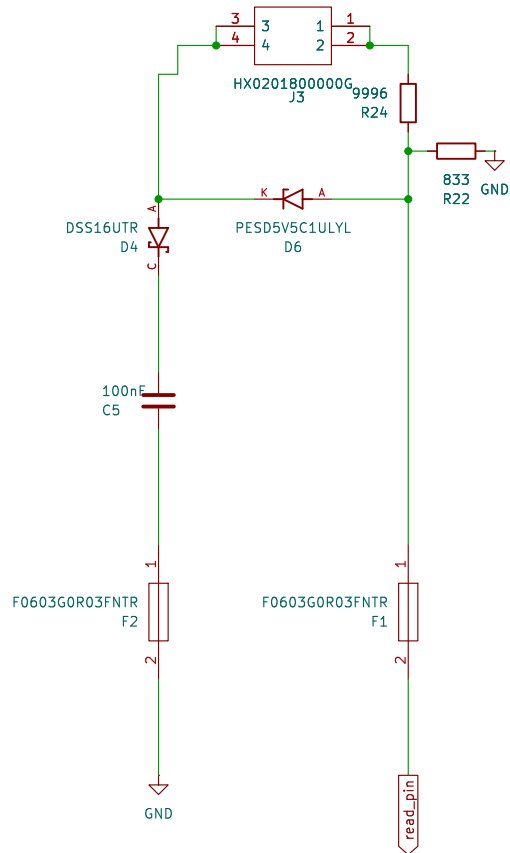
Date: 2024-12-26

Rev: C3I

KiCad E.D.A. 8.0.7

Id: 4/10

## In Julien We Truss Que Ça Fonctionne



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Sheet: / Multimètre/  
File: multimeter.kicad\_sch

**Title: Touillomètre – Multimeter**

Size: A4

Date: 2024-12-26

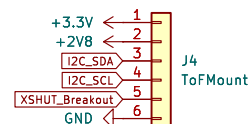
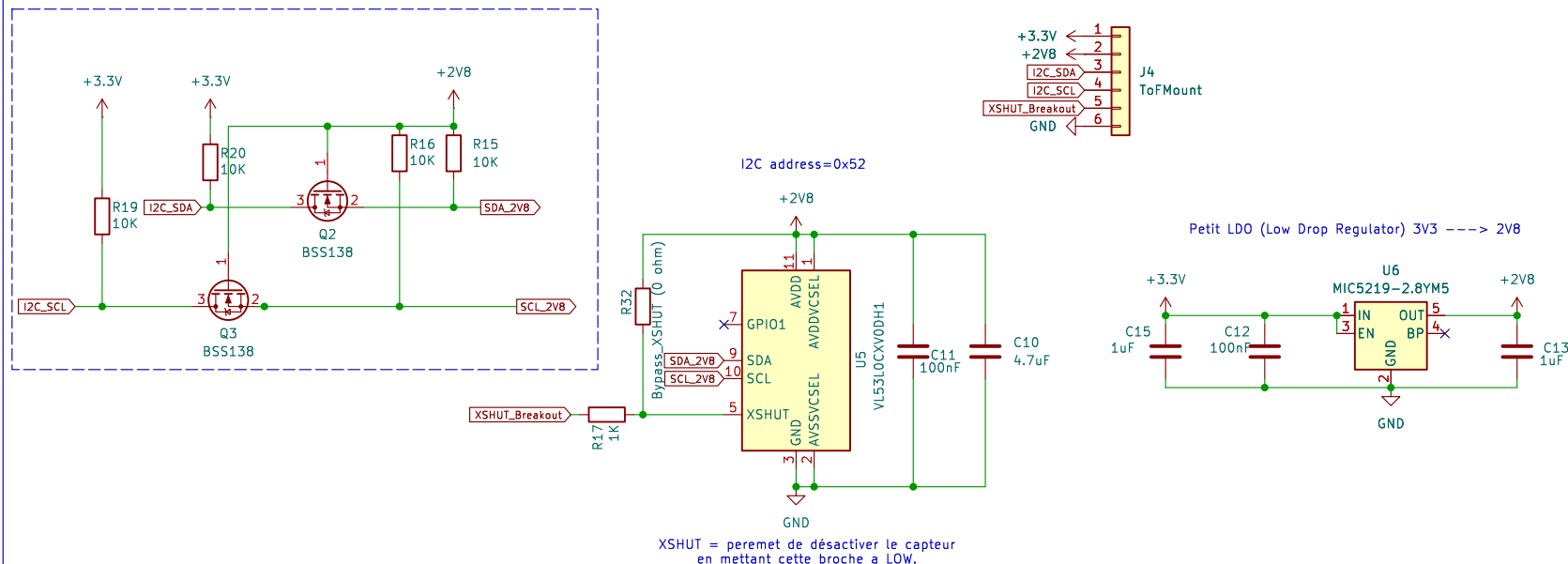
Rev: C3I

|                    |  |
|--------------------|--|
| Size: A1           |  |
| KiCad E.D.A. 8.0.7 |  |

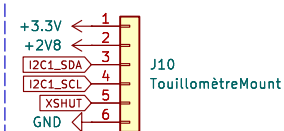
Id: 5/10

**Ceci est un petit PCB  
on veut pouvoir le retirer et le remettre avec un header**

Permet de transformer la logique SDA/SCL qui est de 3.3V a une logique SDA/SCL de 2.8V



Sur le Touillomètre



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Sheet: /distanceSensor (ToF)/  
File: distanceSensor.kicad\_sch

**Title: Touillometre – ToF Distance Sensor**

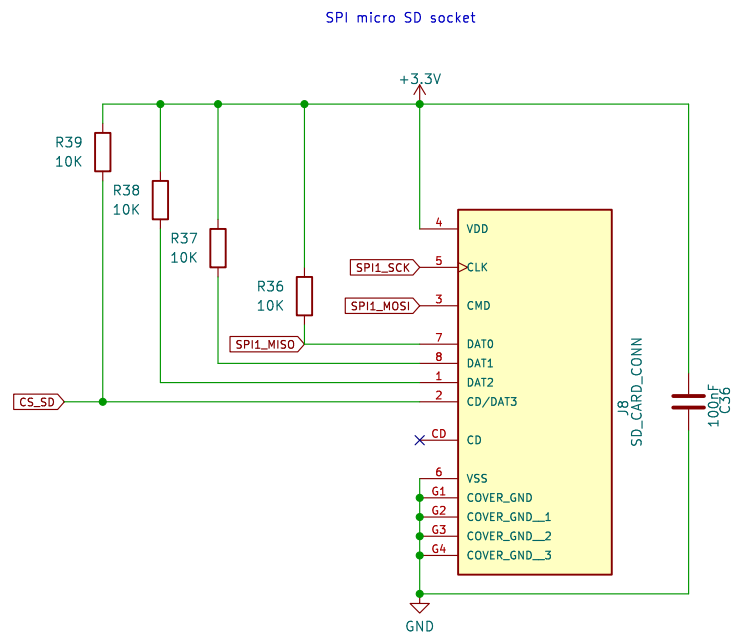
Size: A4

Date: 2024-12-26

Rev: C3I

KiCad E.D.A. 8.0.7

Id: 6/10



University of Sherbrooke

Sheet: /SD card reader (bonus)/

File: SDCardReader.kicad\_sch

**Title: Touillometre – SD Card Reader**

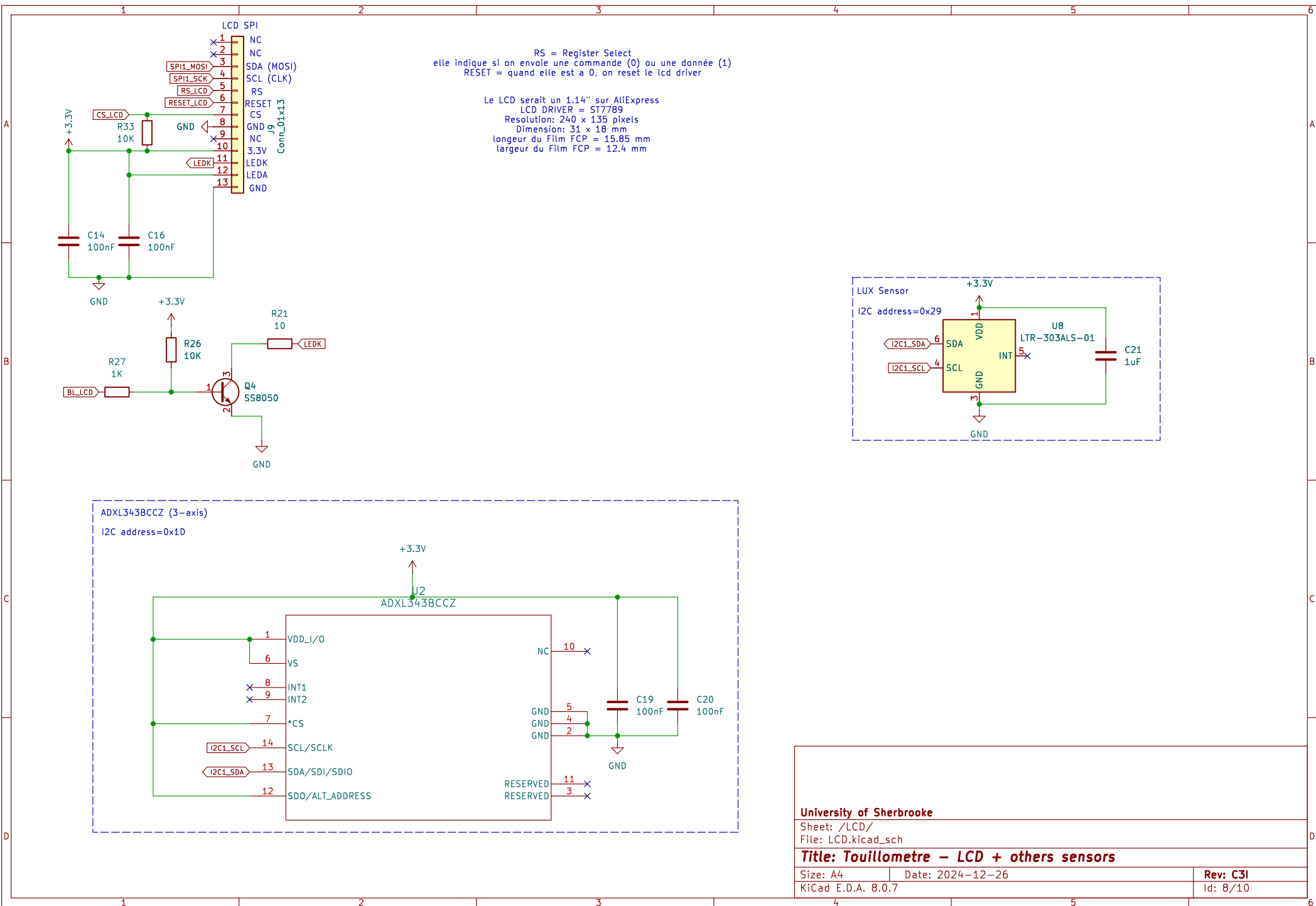
Size: A4

Date: 2024-12-26

Rev: C3I

KiCad E.D.A. 8.0.7

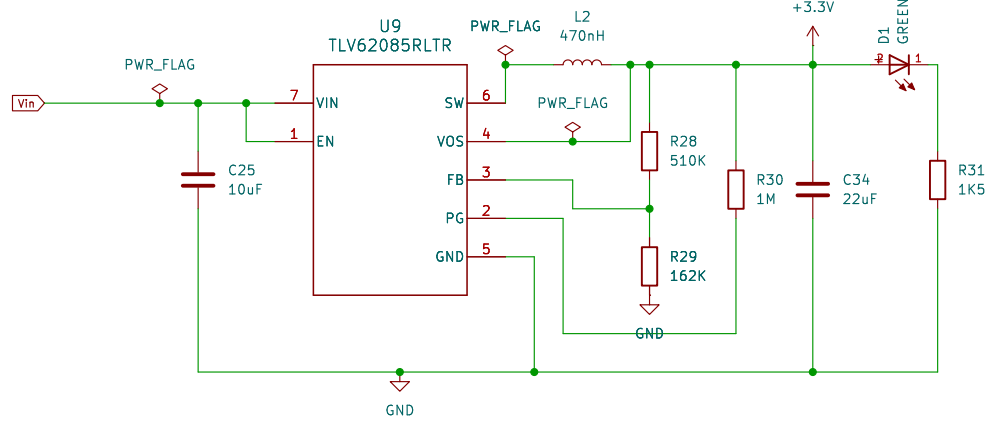
Id: 7/10



# Régulateur tension 3.3V (input: 5V-3.4V)

Vin = 5V (si USB branché)  
Vbat = entre 4.2V et 3.5V

LED QUI INDIQUE QUE LE CIRCUIT EST ALIMENTÉ

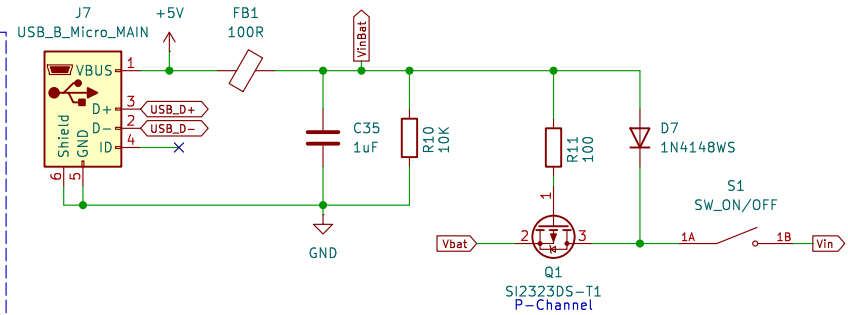
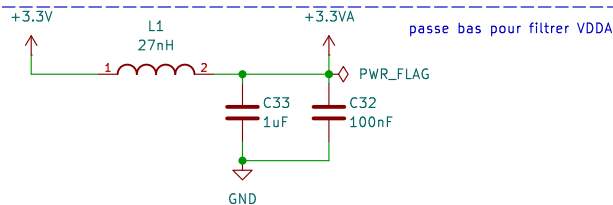
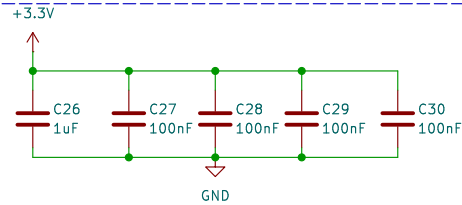


MOSFET N-Channel:  
Si une tension appliqué a la Gate est supérieure que celle a la Source,  
le courant va circuler du Drain a la Source

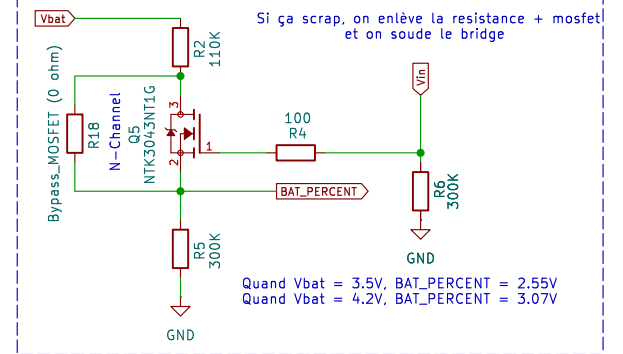
MOSFET P-Channel:  
Si une tension appliqué a la Gate est inférieure a celle de la Source,  
le courant va circuler de la Source au Drain

Découplage entre VDD et VDDA pour réduire le bruit au maximum

NEAR MCU



## Battery percentage charger



## Battery charger management



File: BatteryCharger.kicad\_sch

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Sheet: /MCU and power management/Alim + buck converter/  
File: Alimentation.kicad\_sch

Title: Touillomètre – Power Management

Size: A4

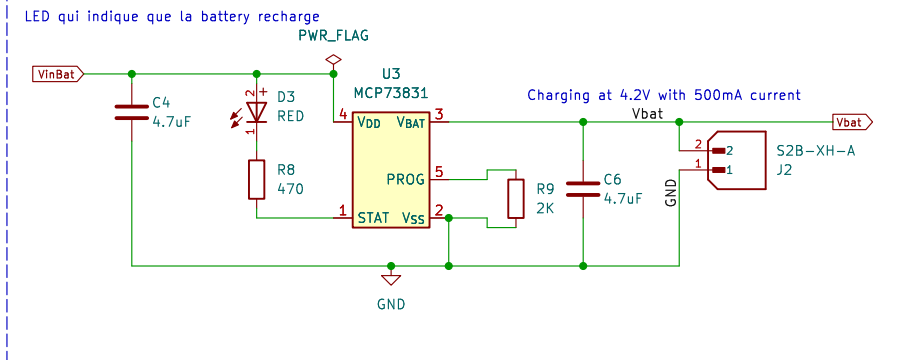
Date: 2024-12-26

Rev: C3I

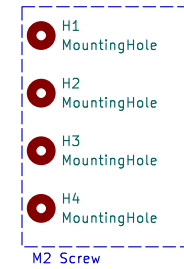
KiCad E.D.A. 8.0.7

Id: 10/10

Charging circuit



Mounting Hole Battery + LCD



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Sheet: /MCU and power management/Alim + buck converter/Battery charger management/  
File: BatteryCharger.kicad\_sch

**Title: Touillomètre – Battery Management**

Size: A4

Date: 2024-12-26

Rev: C3I

KiCad E.D.A. 8.0.7

Id: 11/10